

EcoSort AI

On-device waste segregation and impact analysis on
Snapdragon PCs

Sort smarter. Run local.

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Snapdragon AI Lab Build & Present Challenge



Sorting waste is harder than it looks

1

Wrong bin

One misplaced item can contaminate a whole batch of recycling.

2

Cloud dependence

Most sorting apps need internet and add a round trip for every photo.

3

Privacy

Camera images get uploaded to servers the user never sees.

Sorting needs to be instant, offline, and private.

Point at waste. Get the bin and the impact.

- Classifies an item from a webcam frame or a photo
- Says which bin it goes in, with a confidence score
- Estimates the carbon impact of sorting it correctly
- Adds recycling center map, leaderboard and a chat assistant

Illustrative result card

Category: Plastic

Confidence: shown per item

Bin: recycling

Eco impact score and EPA WARM based estimate

How it works, entirely on the laptop



Local Flask API | React UI | SQLite | EPA WARM emission lookup

No cloud calls. No image leaves the device.

Built for Snapdragon with AI Hub

Compile

MobileNetV3-Large exported to ONNX and compiled for Snapdragon X on AI Hub.

Profile

AI Hub profile job checks per-layer runtime before the model touches my laptop.

Run

ONNX Runtime with the QNN execution provider on the NPU, CPU as fallback.

[FILL] ms

NPU latency per image

[FILL] ms

CPU latency per image

[FILL] MB

Model size

Measured on my Snapdragon laptop with the benchmark script in the repo.

Where it helps

1

Schools and campuses

Teach correct sorting with instant feedback and a class leaderboard.

2

Offices and facilities

Cut contamination at shared bins without extra hardware.

3

Low-connectivity places

Works fully offline after setup, so no signal is needed.

4

Privacy by design

Photos are processed on the device and never uploaded.

What comes next

1

Multi-object detection

Sort several items in one frame.

2

Regional bin rules

Match local recycling regulations.

3

Mobile companion

Sync scores and history from a phone.

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Thank you.